

Electromagnetic Theory

Date

DAY-15 VAO

Applications

- Glucose Meter
- Transformer
- Motor
- IC's
- Wireless charger
- satellites
- laptop / smart watch
- Bullet train

Key Points

- ① Transmission lines
- ② Vector calculus
- ③ Static Electric Field / static magnetic field
- ④ Maxwell's equation

Gauss's law:

$$\phi = \frac{Q}{\epsilon_0}$$

$$\nabla \cdot \vec{E} = \frac{\rho}{\epsilon_0} \quad (DF)$$

$$\oint \vec{E} \cdot \hat{n} \, da = \frac{q_{enc}}{\epsilon_0} \quad (IF)$$

Gauss's law for magnetism:

$$\oint \vec{B} \cdot d\vec{A} = 0 \quad (IF)$$

$$\nabla \cdot \vec{B} = 0 \quad (DF)$$

Ampere's law:

$$\nabla \times \vec{B} = \mu_0 \vec{J} \quad (DF)$$

$$\oint \vec{H} \cdot d\vec{l} = I_{enc} \quad (IF)$$

$$\oint \vec{B} \cdot d\vec{s} = \mu_0 I + \epsilon_0 \mu_0 \frac{d\phi}{dt} \quad (IF)$$

Faraday's law

$$\oint \vec{E} \cdot d\vec{s} = - \frac{d\phi_B}{dt} \quad (IF)$$

$$\nabla \times \vec{E} = - \frac{d\vec{B}}{dt} \quad (DF)$$

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DAY-21 - VEC

Scalar

- Only magnitude
- One dimensional
- scalar can only be divided by another scalar.
- Time, mass, length

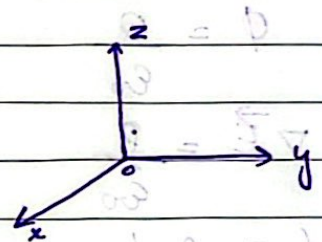
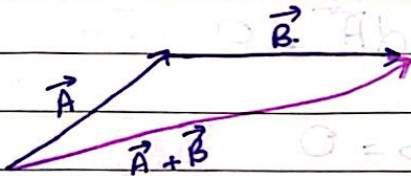
Vector

- Magnitude & direction
- Multi dimensional
- Vector cannot be divided by another vector
- Velocity, force, weight

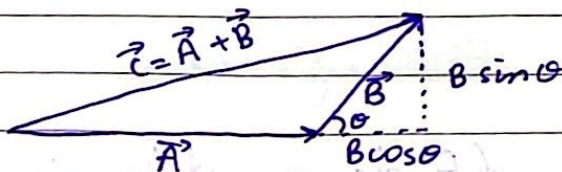
Basics of vector :-

- $\vec{A} = A_x \hat{i} + A_y \hat{j} + A_z \hat{k}$
- $|\vec{A}| = \sqrt{A_x^2 + A_y^2 + A_z^2}$ (magnitude)

$$\hat{A} = \frac{\vec{A}}{|\vec{A}|} = \frac{A_x \hat{i} + A_y \hat{j} + A_z \hat{k}}{\sqrt{A_x^2 + A_y^2 + A_z^2}} = \hat{i}_A \text{ (unit vector)}$$



Example :-



Using pythagorus theorem,

$$C^2 = (A + B \cos \theta)^2 + (B \sin \theta)^2$$

$$C^2 = A^2 + B^2 \cos^2 \theta + 2AB \cos \theta + B^2 \sin^2 \theta$$

$$C^2 = A^2 + 2AB \cos \theta + B^2 (\cos^2 \theta + \sin^2 \theta)$$

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$$C^2 = A^2 + B^2 + 2AB \cos \theta$$

(parallelogram theorem)

Dot product:-

$$C = \vec{A} \cdot \vec{B}$$

$$\vec{A} = A_x \hat{i} + A_y \hat{j} + A_z \hat{k}$$

$$\vec{B} = B_x \hat{i} + B_y \hat{j} + B_z \hat{k}$$

$$\vec{A} \cdot \vec{B} = |A| |B| \cos \theta$$

resultant is a scalar quantity

$$= A_x B_x + A_y B_y + A_z B_z$$

Cross Product:-

$$\vec{A} \times \vec{B} = |A| |B| \sin \theta$$

$$\vec{C} = \vec{A} \times \vec{B} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix}$$

resultant is a vector quantity

Example:-

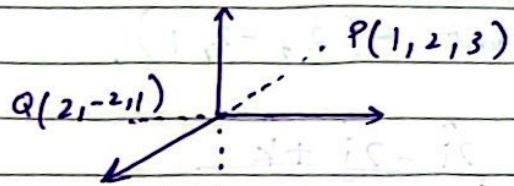
$$\vec{P} = 1\hat{i} + 2\hat{j} + 3\hat{k}$$

$$\vec{Q} = 2\hat{i} - 2\hat{j} + \hat{k}$$

$$\vec{PQ} = \vec{OQ} - \vec{OP}$$

$$= 2\hat{i} - 2\hat{j} + \hat{k} - 1\hat{i} - 2\hat{j} - 3\hat{k}$$

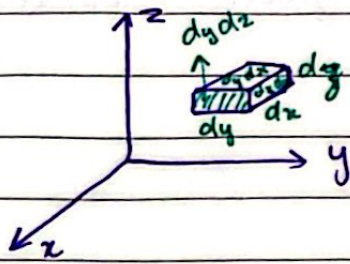
$$\vec{PQ} = \hat{i} - 4\hat{j} - 2\hat{k}$$



$$|\vec{PQ}| = \sqrt{(1)^2 + (-4)^2 + (-2)^2}$$
$$= \sqrt{1 + 16 + 4}$$

$$|\vec{PQ}| = \sqrt{21}$$

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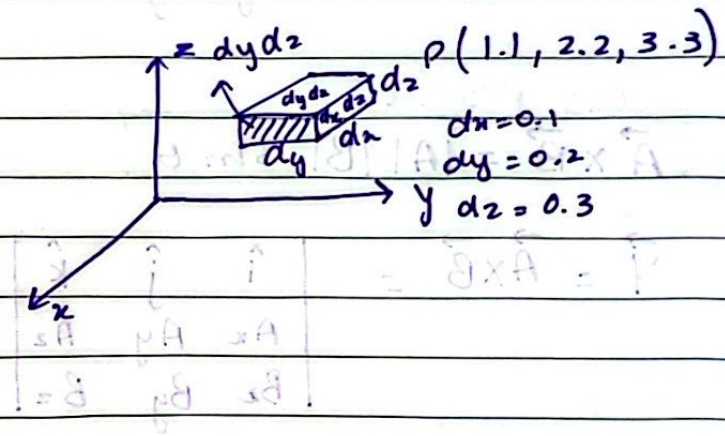
$$dv = dx dy dz$$

$$|dl| = \sqrt{dx^2 + dy^2 + dz^2}$$

Example:-

$$dv = dx dy dz$$

$$= 0.06$$



Example:-

Specify the unit vector from the origin towards the point $F(2, -2, 1)$.

$$\vec{F} = 2\hat{i} - 2\hat{j} + \hat{k}$$

$$|F| = \sqrt{2^2 + (-2)^2 + (1)^2}$$

$$= \sqrt{9}$$

$$\hat{F} = \frac{\vec{F}}{|F|}$$

$$= \frac{2\hat{i} - 2\hat{j} + \hat{k}}{3}$$

$$\hat{i}_F = \frac{2}{3}\hat{i} - \frac{2}{3}\hat{j} + \frac{1}{3}\hat{k}$$

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Example:-

$$\vec{U}(-1, 2, 1), \vec{L}(3, -3, 0), \vec{S}(-2, -3, -4)$$

$$\vec{R}_{UL} = ?, \vec{R}_{UL} + \vec{R}_{US} = ?, |\vec{r}_U| = ?, |\vec{r}_S| = ?$$

$$\text{unit vector of } \vec{R}_{US} = ?$$

$$\vec{r}_U = -1\hat{i} + 2\hat{j} + \hat{k}$$

$$|\vec{r}_U| = \sqrt{(-1)^2 + (2)^2 + (1)^2}$$
$$= \sqrt{4}$$

$$|\vec{r}_U| = 2$$

$$\vec{r}_L = 3\hat{i} - 3\hat{j}$$

$$\vec{R}_{UL} = \vec{r}_L - \vec{r}_U$$

$$= 3\hat{i} - 3\hat{j} + \hat{i} - 2\hat{j} - \hat{k}$$

$$= 4\hat{i} - 5\hat{j} - \hat{k}$$

$$\vec{r}_S = -2\hat{i} - 3\hat{j} - 4\hat{k}$$

$$|\vec{r}_S| = \sqrt{(-2)^2 + (-3)^2 + (-4)^2}$$
$$= \sqrt{4 + 9 + 16}$$

$$|\vec{r}_S| = \sqrt{29}$$

$$\vec{R}_{US} = \vec{r}_S - \vec{r}_U$$

$$= -2\hat{i} - 3\hat{j} - 4\hat{k} + \hat{i} - 2\hat{j} - \hat{k}$$

$$= -\hat{i} - 5\hat{j} - 5\hat{k}$$

$$\vec{R}_{UL} + \vec{R}_{US} = 4\hat{i} - 5\hat{j} - \hat{k} - \hat{i} - 5\hat{j} - 5\hat{k}$$

$$= 3\hat{i} - 10\hat{j} - 6\hat{k}$$

$$|\vec{R}_{US}| = \sqrt{(1)^2 + (5)^2 + (5)^2}$$

$$= \sqrt{1 + 25 + 25}$$

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$$|R_{us}| = \sqrt{51}$$

$$\hat{R}_{us} = \frac{\vec{R}_{us}}{|R_{us}|}$$
$$= \frac{-\hat{i} - 5\hat{j} - 5\hat{k}}{\sqrt{51}}$$

$$\hat{R}_{us} = -\frac{1}{\sqrt{51}}\hat{i} - \frac{5}{\sqrt{51}}\hat{j} - \frac{5}{\sqrt{51}}\hat{k}$$

Date

$$|R_{us}| = \sqrt{51}$$

$$\hat{R}_{us} = \frac{\vec{R}_{us}}{|R_{us}|}$$
$$= \frac{-\hat{i} - 5\hat{j} - 5\hat{k}}{\sqrt{51}}$$

$$\hat{R}_{us} = -\frac{1}{\sqrt{51}}\hat{i} - \frac{5}{\sqrt{51}}\hat{j} - \frac{5}{\sqrt{51}}\hat{k}$$

DAY-3

Cylindrical Co-ordinate System:-

Real World Applications:-

- Electromagnetics
- Fluid dynamics
- Electromagnetic field in wires
- Flow in pipes
- Heat conduction in cylinders

Cylindrical C.S

- Defined by ρ , ϕ and z .
- ρ (radial distance from the z -axis)

ϕ (Angle it makes in the xy plane)

z (distance along z axis)

Rectangular C.S

- Defined by x , y and z -axis

$$\hat{x} - \hat{y} - \hat{z} + \hat{x} - \hat{y} - \hat{z} =$$
$$\hat{x} - \hat{y} - \hat{z} - \hat{x} - \hat{y} - \hat{z} =$$

$$\hat{x} - \hat{y} - \hat{z} - \hat{x} - \hat{y} - \hat{z} = 2\hat{x} + 2\hat{y} + 2\hat{z}$$
$$\hat{x} - \hat{y} - \hat{z} - \hat{x} - \hat{y} - \hat{z} =$$

Cylindrical Coordinates:-

- Surfaces of constant ρ are centered along z -axis

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- Surfaces of constant ϕ are planes passing through z-axis.

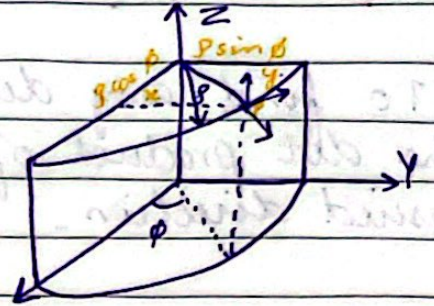
Surfaces of constant z are horizontal planes p .

Formulas:-

$$p = \sqrt{x^2 + y^2}$$

$$\tan \phi = \frac{y}{x} \Rightarrow \phi = \tan^{-1}\left(\frac{y}{x}\right)$$

$$z = z$$



Conversion from CCS to RCS p, ϕ, z given

$$x = p \cos \phi$$

$$y = p \sin \phi$$

$$z = z$$

Unit vectors :-

• a_p, a_ϕ, a_z (CCS)

• a_x, a_y, a_z (RCS)

Transforming Unit Vectors

Dot Product

	a_p	a_ϕ	a_z
a_x	$\cos \phi$	$-\sin \phi$	0
a_y	$\sin \phi$	$\cos \phi$	0
a_z	0	0	1

$$A = A_p a_p + A_\phi a_\phi + A_z a_z$$

$$A = A_x a_x + A_y a_y + A_z a_z$$

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$$A_{\text{rect}} = A_x a_x + A_y a_y + A_z a_z$$

$$A_{\text{cyl}} = A_\rho a_\rho + A_\phi a_\phi + A_z a_z$$

$$A_\rho = ? \quad A_\phi = ? \quad A_z = ?$$

To find any desired component of a vector, we take the dot product of the vector and a unit vector in desired direction.

$$A_\rho = A_{\text{rect}} \cdot a_\rho = (A_x a_x + A_y a_y + A_z a_z) \cdot a_\rho$$

$$A_\phi = A_{\text{rect}} \cdot a_\phi$$

$$A_z = A_{\text{rect}} \cdot a_z$$

$$= A_x (a_x \cdot a_\rho) + A_y (a_y \cdot a_\rho) + A_z (a_z \cdot a_\rho)$$

$$= A_x \cos \phi + A_y (\sin \phi) + A_z (0)$$

$$A_\rho = A_x \cos \phi + A_y \sin \phi$$

$$A_\phi = A_{\text{rect}} \cdot a_\phi$$

$$= (A_x a_x + A_y a_y + A_z a_z) \cdot a_\phi$$

$$= A_x (a_x \cdot a_\phi) + A_y (a_y \cdot a_\phi) + A_z (a_z \cdot a_\phi)$$

$$= A_x (-\sin \phi) + A_y (\cos \phi) + A_z (0)$$

$$A_\phi = -A_x \sin \phi + A_y \cos \phi$$

$$A_z = A_{\text{rect}} \cdot a_z$$

$$= (A_x a_x + A_y a_y + A_z a_z) \cdot a_z$$

$$= A_x (a_x \cdot a_z) + A_y (a_y \cdot a_z) + A_z (a_z \cdot a_z)$$

$$= A_x (0) + A_y (0) + A_z (1)$$

$$A_z = A_z$$

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$$B = y a_x - x a_y - z a_z$$

Transform it into cylindrical coordinates

$$B_{cyl} = B_r a_r + B_\phi a_\phi + B_z a_z$$

$$B_r = B \cdot a_r$$

$$= (y a_x - x a_y - z a_z) \cdot a_r$$

$$= y(a_x \cdot a_r) - x(a_y \cdot a_r) - z(a_z \cdot a_r)$$

$$= y(\cos \phi) - x(\sin \phi) - z(0)$$

$$= y \cos \phi - x \sin \phi$$

$$B_\phi = B \cdot a_\phi$$

$$= (y a_x - x a_y - z a_z) \cdot a_\phi$$

$$= -y \sin \phi - x \cos \phi$$

$$\phi = \tan^{-1}\left(\frac{x}{y}\right)$$

DAY-N

$$\rho = \sqrt{(-y)^2 + (-x)^2}$$

Example 1.3

→ HW

D#1.5 & 1.6

HOME WORK:-

Transform $B = y a_x - x a_y + z a_z$ into cylindrical co-ordinates

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$$B_{cyl} = B_r a_r + B_\phi a_\phi + B_z a_z$$

$$B_r = (y a_x - x a_y + z a_z) \cdot a_r$$

$$= y(a_x \cdot a_r) - x(a_y \cdot a_r) + z(a_z \cdot a_r)$$

$$= y \cos \phi - x \sin \phi + z(0)$$

$$= y \cos \phi - x \sin \phi$$

putting value of 'y' and 'x'

$$= p \sin \phi \cos \phi - p \cos \phi \sin \phi$$

$$B_p = 0$$

$$B_\phi = (y a_x - x a_y + z a_z) \cdot a_\phi$$

$$= y(a_x a_\phi) - x(a_y a_\phi) + z(a_z a_\phi)$$

$$= y(-\sin \phi) - x(\cos \phi) + z(0)$$

$$= -y \sin \phi - x \cos \phi$$

putting the value of x and y

$$= -p \sin^2 \phi - p \cos^2 \phi$$

$$= -p(\sin^2 \phi + \cos^2 \phi)$$

$$= -p(1)$$

$$B_\phi = -p$$

DAY-4

Spherical Co-ordinate System :-

$r \rightarrow$ The distance from origin to the point

RCS $\rightarrow r, \theta, \phi$

Polar Angle (θ) $\rightarrow$ The angle b/w the z-axis and the line connecting to the origin.

CCS $\rightarrow r, \theta, \phi$

SCS $\rightarrow r, \phi, \theta$

Azimuthal Angle (ϕ) $\rightarrow$ The angle b/w the x-axis and projection of the line connecting the origin to the point onto the xy-plane.

Unit Vector :-

$a_r \rightarrow$ Radial unit vector, directed outward

RCS $\rightarrow$ 3 planes

CCS $\rightarrow$ 2 planes, 1 angle

SCS $\rightarrow$ 1 plane, 2 angles

from the origin, normal to the sphere

$a_\theta \rightarrow$ Unit vector, tangent to the surface and pointing in

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the direction of increasing θ , normal to the conical sphere.

$a_\phi \rightarrow$ Unit vector, tangent to the sphere and cone, pointing in the direction of increasing ϕ .

Transformation from RCS to SCS

$$(x, y, z) \rightarrow (r, \theta, \phi)$$

$$z = r \cos \theta$$

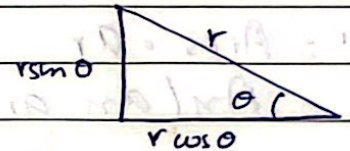
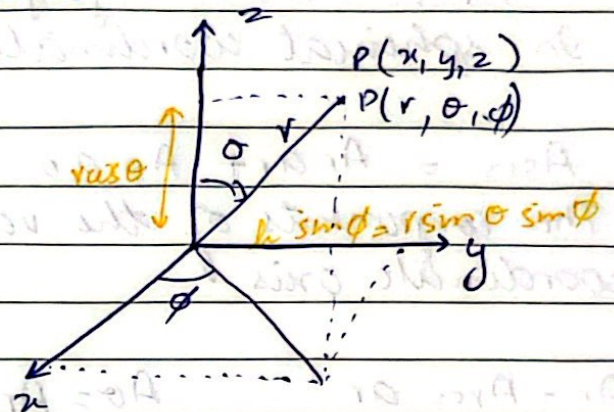
$$x = r \sin \theta \cos \phi$$

$$y = r \sin \theta \sin \phi$$

$$r = \sqrt{x^2 + y^2 + z^2}$$

$$\phi = \tan^{-1} \left(\frac{y}{x} \right)$$

$$\cos \theta = \frac{z}{r} = \frac{z}{\sqrt{x^2 + y^2 + z^2}} \Rightarrow \theta = \cos^{-1} \left(\frac{z}{\sqrt{x^2 + y^2 + z^2}} \right)$$



Transformation of vectors :-

The transformation of vectors requires determining the dot products of spherical unit vectors with rectangular unit vectors.

Purpose of dot product :-

To find the component of a vector in the

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direction of a new unit vector (from SCS), we take the dot product of original vectors (from RCS) with the new vector.

Mathematical Representation :-

$$A_{RCS} = A_x a_x + A_y a_y + A_z a_z$$

In spherical coordinates system

$$A_{SCS} = A_r a_r + A_\theta a_\theta + A_\phi a_\phi$$

The components of the vector A_{RCS} along the spherical coordinate axis

$$A_r = A_{RCS} a_r$$

$$A_\theta = A_{RCS} a_\theta$$

$$A_\phi = A_{RCS} a_\phi$$

$$A_r = A_{RCS} \cdot a_r$$

$$= A_x (a_x \cdot a_r) + A_y (a_y \cdot a_r) + A_z (a_z \cdot a_r)$$



Dot products - that give the projections of the rectangular unit vector onto the spherical unit vector ' a_r '

$$A_\theta = A_{RCS} a_\theta$$

$$= A_x (a_x \cdot a_\theta) + A_y (a_y \cdot a_\theta) + A_z (a_z \cdot a_\theta)$$

$$A_\phi = A_{RCS} a_\phi$$

$$= A_x (a_x \cdot a_\phi) + A_y (a_y \cdot a_\phi) + A_z (a_z \cdot a_\phi)$$

HOMEWORK:

D 1.5

a) Given:

$$\rho = 4.4$$

$$\phi = -115^\circ$$

$$z = 2$$

sol.

$$x = \rho \cos \phi$$

$$x = 4.4 \cos(-115^\circ)$$

$$x = -1.860$$

$$y = \rho \sin \phi$$

$$= 4.4 \sin(-115^\circ)$$

$$y = -3.99$$

$$z = z$$

$$z = 2$$

b) Given:

$$x = -3.1$$

$$y = 2.6$$

$$z = -3$$

sol:

$$\rho = \sqrt{x^2 + y^2}$$

$$= \sqrt{(-3.1)^2 + (2.6)^2}$$

$$\rho = 4.05$$

$$\phi = \tan^{-1} \frac{y}{x}$$

$$\phi = \tan^{-1} \left(\frac{2.6}{-3.1} \right)$$

$$\phi = 140.0^\circ$$

(As it was in the 2nd quad, so, we subtract the angle with 180.)

$$z = -yaz = z$$

$$z = -3$$

c) Distance CD = $\sqrt{D-C}$

$$= \sqrt{(-3.1 + 1.860)^2 + (2.6 + 3.99)^2 + (-3 - 2)^2}$$

$$= 8.36$$

D 1.6

$$F = 10ax - 8ay + 6az$$

$$F_{cyl} = F_{\rho} a_{\rho} + F_{\phi} a_{\phi} + F_z a_z$$

$$F_{\rho} = (10ax - 8ay + 6az) a_{\rho}$$

$$= 10(a_x a_{\rho}) - 8(a_y a_{\rho}) + 6(a_z a_{\rho})$$

$$= 10 \cos \phi - 8 \sin \phi$$

$$F_{\phi} = (10ax - 8ay + 6az) a_{\phi}$$

$$= 10(a_x a_{\phi}) - 8(a_y a_{\phi}) + 6(a_z a_{\phi})$$

$$= 10(-\sin \phi) - 8(\cos \phi) + 0$$

$$= -10 \sin \phi - 8 \cos \phi$$

$$F_z = 6$$

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$$F_{cyl} = (10 \cos \phi - 8 \sin \phi) a_\rho + (-\overset{10 \sin}{\cos \phi} - 8 \cos \phi) a_\phi + 6(a_z)$$

$$\phi = \tan^{-1} \left(\frac{y}{x} \right)$$

$$F_{cyl} = 12.806 a_\rho + 6 a_z$$

DAY-5

Transformation of Vectors:

$$A_{rcs} = A_x a_x + A_y a_y + A_z a_z$$

$$A_{scs} = A_r a_r + A_\theta a_\theta + A_\phi a_\phi$$

To find the component of a vector in the direction of new unit vector (from scs), we take the dot product of the original vector's unit vector with the new vector.

$$A_r = A_{rcs} a_r = A_x (a_x a_r) + A_y (a_y a_r) + A_z (a_z a_r)$$

$$A_r = A_x \cos \phi \sin \theta + A_y \sin \phi \sin \theta + A_z \cos \theta$$

a_r = points radially outward from the origin.

a_θ = Tangent to spherical surface and pointing in the direction of increasing θ

a_ϕ = Tangent to the circle of constant 'r' and ' θ ' pointing in the direction of increasing ϕ .

$$a_x a_r = a_r a_x = \cos \phi \sin \theta$$

$$a_y a_r = a_r a_y = \sin \phi \sin \theta$$

$$a_z a_r = a_r a_z = \cos \theta$$

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	a_r	a_θ	a_ϕ
a_x	$\sin \theta \cos \phi$	$\cos \theta \cos \phi$	$-\sin \phi$
a_y	$\sin \theta \sin \phi$	$\cos \theta \sin \phi$	$+\cos \phi$
a_z	$\cos \theta$	$-\sin \theta$	0

$$A_\theta = \text{Arcs } a_\theta = A_x (a_x a_\theta) + A_y (a_y a_\theta) + A_z (a_z a_\theta)$$

$$A_\theta = A_x \cos \theta \cos \phi + A_y \cos \theta \sin \phi + A_z \sin \theta$$

$$A_\phi = \text{Arcs } a_\phi = A_x (a_x a_\phi) + A_y (a_y a_\phi) + A_z (a_z a_\phi)$$

$$A_\phi = -A_x \sin \theta + A_y \cos \theta + 0$$

QUESTION:-

$G = \frac{xz}{y} a_x \rightarrow$ express it in spherical coordinate system

step 1:-

$$G = G_r a_r + G_\theta a_\theta + G_\phi a_\phi$$

$$G_r = G \cdot a_r = \frac{xz}{y} a_x \cdot a_r \quad (\text{step 2:-})$$

$$= \frac{(r \sin \theta \cos \phi)(r \cos \theta)}{(r \sin \theta \sin \phi)} (a_x \cdot a_r)$$

$$= \frac{r \cos \theta \cos \phi}{\sin \phi} (a_x \cdot a_r)$$

$$= \frac{r \cos \theta \cos \phi}{\sin \phi} / \sin \theta \cos \phi$$

$$G_r = \frac{r \cos^2 \phi \cos \theta}{\sin \phi} \sin \theta$$

step 3:-

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$$G_\theta = G_{a\theta} = \frac{xz}{y} \cdot \frac{a_\theta}{a_\theta} = \frac{xz}{y}$$

$$= \frac{(r \cos \phi \sin \theta)(r \cos \theta)}{r \sin \theta} (\cos \theta \cos \phi)$$

$$G_\theta = r \cos^2 \theta \cos^2 \phi$$

$$G_\phi = G_{a\phi} = \frac{xz}{y} \cdot \frac{a_\phi}{a_\phi} = \frac{xz}{y}$$

$$G_\phi = r \cos \theta \cos \phi$$

$$G = G_r a_r + G_\theta a_\theta + G_\phi a_\phi$$

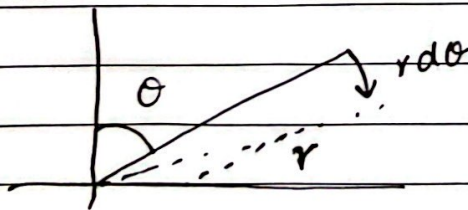
$$G = \frac{r \cos \theta \cos^2 \phi \sin \theta}{\sin \phi} a_r + \frac{r \cos^2 \theta \cos^2 \phi}{\sin \phi} a_\theta - r \cos \theta \cos \phi a_\phi$$

DAY-6

Differential Elements in the spherical Coordinate System:

Differential lengths:-

- Consider increasing r , θ and ϕ by small amounts $\rightarrow dr$, $d\theta$ and $d\phi$ respectively
- $dr \rightarrow$ along the radial direction (r)
- $r d\theta \rightarrow$ along the polar direction (θ)
- $r \sin \theta d\phi \rightarrow$ along the azimuthal direction (ϕ)



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Differential Surface Area:

surface area spanned by r and θ $\vec{dr} \times r d\theta = r dr d\theta$

surface area spanned by r and ϕ $\vec{dr} \times (r \sin \theta d\phi) = r \sin \theta dr d\phi$

surface area spanned by θ and ϕ $\vec{r} d\theta \times (r \sin \theta d\phi) = r^2 \sin \theta d\theta d\phi$

Differential volume:

$$= dr \times r d\theta \times r \sin \theta d\phi = r^2 \sin \theta dr d\theta d\phi = dv$$

Solve:-

$$\int dv = \int r^2 \sin \theta dr d\theta d\phi$$

$$V_r = \frac{r^3}{3} \sin \theta d\theta d\phi \quad (\text{w.r.t 'r'})$$

$$V_\theta = -r^2 \cos \theta dr d\phi \quad (\text{w.r.t '}\theta\text{'})$$

$$V_\phi = \phi r^2 \sin \theta dr d\theta \quad (\text{w.r.t '}\phi\text{'})$$

$$V = \int_\phi \int_\theta \int_r dv = -\frac{r^3}{3} \cos \theta \phi$$

QUESTION:-

→ $C(-3, 2, 1) \rightarrow$ spherical coordinates?

Sol:-

$$x = -3, y = 2, z = 1$$

$$\begin{aligned} r &= \sqrt{(-3)^2 + (2)^2 + (1)^2} \\ &= \sqrt{9 + 4 + 1} \\ &= \sqrt{14} \end{aligned}$$

$$\theta = \tan^{-1} \left(\frac{y}{x} \right)$$

Date

$$\phi = \tan^{-1} \left(\frac{2}{-3} \right) \Rightarrow 146.30^\circ$$

$$\theta = \cos^{-1} \left(\frac{1}{\sqrt{14}} \right) = 74.49^\circ$$

$$\rightarrow D(r=5, \theta=20^\circ, \phi=70^\circ)$$

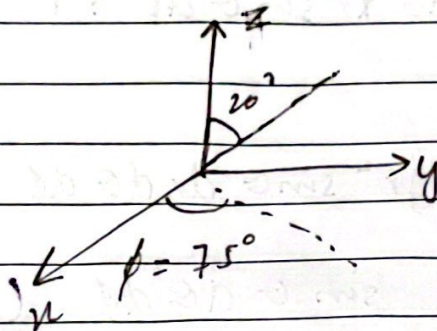
Sol:-

$$x = r \sin \theta \cos \phi$$

$$x = 0.5$$

$$y = r \sin \theta \sin \phi$$
$$= -1.66$$

$$z = r \cos \theta$$
$$= 4.69$$



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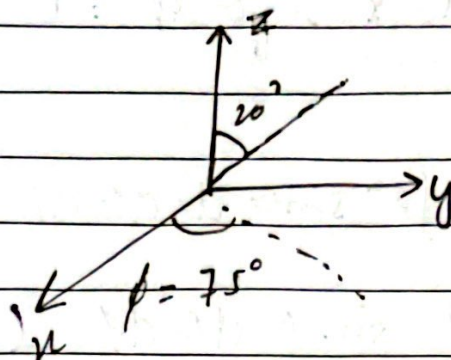
$$x = 0.5$$

$$y = r \sin \theta \sin \phi$$

$$= -1.66$$

$$z = r \cos \theta$$

$$= 4.69$$



DAY-7

Coulomb's Law:-

Force between the two charged particles is: $F = K \frac{Q_1 Q_2}{R^2}$

- Proportional to the magnitude of each charge (Q_1 and Q_2)
- Inversely proportional to the square of the distance b/w the two charges (R).
- Acts along the line joining the two charges.
- Is repulsive if the charges are of same sign and attractive if they are of opposite signs.

Date

- Q_1 and Q_2 are the magnitudes of the charges
- R is the distance b/w the charges.
- K is the constant of proportionality

$$K = \frac{1}{4\pi\epsilon_0}$$

- ϵ_0 is the permittivity of free space

$$\epsilon_0 = 8.854 \times 10^{-12} \text{ F/m}$$

Components of Coulomb's Law:-

1. Permittivity of free space (ϵ_0) $\rightarrow$ crucial for determining how electric fields interact in a vacuum. It is not dimensionless and has units $\text{C}^2/\text{N}\cdot\text{m}^2$ or F/m .
2. The Coulomb (C) :- It is the SI unit of electric charge. The charge of a single electron is $-1.602 \times 10^{-19} \text{ C}$. 1 Coulomb corresponds to 6×10^{18} electrons.
3. Force magnitude :- Force between two charges of 1C, separated by 1m in a vacuum.

Date

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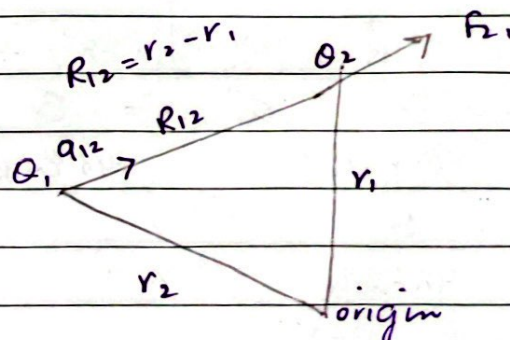
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DAY-8

Coulomb's Law:- (Vector Form)

$$F_2 = \frac{Q_1 Q_2}{4\pi\epsilon_0 R_{12}^2} \hat{a}_{12}$$

$$\hat{a}_{12} = \frac{\vec{R}_{12}}{|\vec{R}_{12}|} = \frac{\vec{r}_2 - \vec{r}_1}{|r_2 - r_1|}$$



Date

Consider a fixed charge Q_1 and test charge Q_2 is moved around Q_1 .

- $\hat{a}_{12}$ is the unit vector pointing from Q_1 to Q_2

Electric field intensity is force experienced by a unit positive charge placed at a point in the field due to charge Q_1 .

$$F_e = \frac{Q_1 Q_2}{4\pi\epsilon_0 R_{12}^2} \hat{a}_{12}$$

$$E_e = \frac{F_e}{Q_2} = \frac{Q_1}{4\pi\epsilon_0 R^2} \hat{a}_{12}$$

Electric Field of a Point charge:-

$$E = \frac{Q}{4\pi\epsilon_0 R^2} \hat{a}_R \rightarrow \hat{a}_R \text{ is the unit vector pointing from the charge } Q \text{ to the point where } E \text{ is measured.}$$

If the charge is placed at the origin of SCS, the unit vector $\hat{a}_R$ becomes the radial unit vector $\hat{a}_R$ and distance R becomes r .

Electric Field in a Rectangular Coordinate system:-

Suppose the charge is located at position $\vec{r}' = x'a_x + y'a_y + z'a_z$. We want to find the electric field intensity at a point $\vec{r} = x\hat{a}_x + y\hat{a}_y + z\hat{a}_z$. The distance b/w charge and the point of interest is $\vec{R} = \vec{r} - \vec{r}'$.

$$\text{Magnitude of } \vec{R} \text{ is } |\vec{R}| = \sqrt{(x-x')^2 + (y-y')^2 + (z-z')^2}$$

$$E = \frac{Q}{4\pi\epsilon_0 R^2} \hat{a}_{\vec{r}-\vec{r}'} = \frac{Q}{4\pi\epsilon_0 R^2} \frac{\vec{r}-\vec{r}'}{|\vec{r}-\vec{r}'|}$$

$$= \frac{Q (\vec{r} - \vec{r}')}{4\pi\epsilon_0 (|\vec{r} - \vec{r}'|^2 |\vec{r} - \vec{r}'|)}$$

$$= \frac{Q (\vec{r} - \vec{r}')}{4\pi\epsilon_0 (|\vec{r} - \vec{r}'|^3)}$$

Date

Question

$$Q_1 = 4 \mu\text{C}, Q_2 = -3 \mu\text{C}$$

$$(2, 0, 0) \quad (0, 2, 0)$$

calculate total E field at $(1, 1, 0)$

a) Distance from Q_1 to $(1, 1, 0)$

$$|\vec{r} - \vec{r}_1|$$

$$E_1 = \frac{4 \times 10^{-6}}{4\pi\epsilon_0 r^2}$$

DAY-9

Volume Charge Density :-

$\rho_v \rightarrow$ Describes how much charge is distributed in a small volume of space

$\text{C/m}^3 \rightarrow$ Coulomb's per cubic meter

$$\rho_v = \frac{\text{Total charge}}{\text{Total volume}} = \frac{dQ}{dv}$$

$$dQ = \rho_v dv$$

$$Q = \int_V \rho_v dv$$

$$= \iiint_{dz dy dx} \rho_v dx dy dz \rightarrow \text{RCS}$$

$$= \int_{dz} \int_{d\phi} \int_{dp} \rho_v dp d\phi dz \rightarrow \text{CCS}$$

$$= \iiint_{d\phi d\theta dr} \rho dr d\theta d\phi \rightarrow \text{SCS}$$

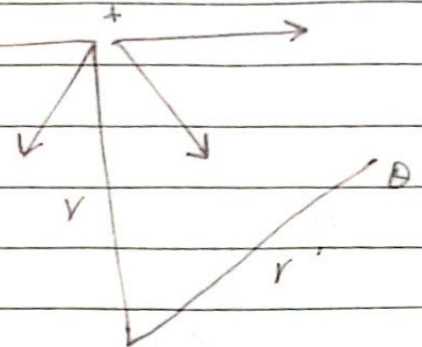
Date

small charge
↑

$$dE(r) = dQ \cdot \frac{r-r'}{|r-r'|^3}$$

$$E(r) = \int \frac{\rho_v dv}{4\pi\epsilon_0 |r-r'|^3} \cdot \frac{r-r'}{|r-r'|}$$

If source point is +ve, its direction is outward.



Total charge contained in a 2cm length of an electron beam.

ρ_v - VAO

$$\rho_v = -5 \times 10^{-6} \text{ cm/m}^3$$

$$Q = \int_V \rho_v dv$$

$$= \int_{0.01}^{0.04} \int_0^{2\pi} \int_0^{0.01} -5 \times 10^{-6} \rho \, dp \, d\phi \, dz$$

$E(r) = ?$ if source point is located at $(0,1,0)$
and $E(r)$ needs to be found at $(0,0,1)$

We needed to solve!

Date

Line Charge:-

$$\rho_L = \frac{\text{total charge}}{\text{Total length}} = \frac{dQ}{dL}$$

Date

Line charge:-

$$\rho_L = \frac{\text{total charge}}{\text{Total length}} = \frac{dQ}{dL}$$

DAY-10

(After MID-1)

Field of a Sheet Charge:-

Surface Charge Density $\rightarrow$ Amount of electric charge per unit area of a surface. (C/m^2)

Infinite sheet of charge $\rightarrow$ Let's place an infinite of charge in the yz plane with uniform charge density ρ_s .

Due to the symmetry, the electric field cannot vary along y -axis or z -axis because any point in the sheet looks identical along these direction

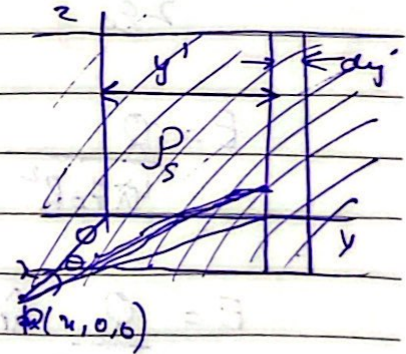
The line charge density of the thin strip (in infinite sheet of charge)

$$\rho_L = \rho_s dy'$$

$$R = \sqrt{x^2 + y'^2}$$

Using Coulomb's law, the differential contribution to the electric field dE_x from the strip is:

$$dE_x = \frac{\rho_L x}{2\pi\epsilon_0 R^2} = \frac{\rho_s dy' x}{2\pi\epsilon_0 (x^2 + y'^2)}$$



Integrating over all the differential strips along the y -axis gives the total field at E_x at point P . **11-VAD**

$$E_x = \int_{-\infty}^{\infty} dE_x = \int_{-\infty}^{\infty} \frac{\rho_s dy' x}{2\pi\epsilon_0 (x^2 + y'^2)} = \frac{\rho_s x}{2\pi\epsilon_0} \int_{-\infty}^{\infty} \frac{dy'}{(x^2 + y'^2)}$$

Date

$$E_x = \frac{\rho_s}{2\epsilon_0}$$

$$\int_{-\infty}^{\infty} \frac{dy'}{x^2 + y'^2} = \frac{1}{x} \tan^{-1}\left(\frac{y'}{x}\right) \Big|_{-\infty}^{\infty}$$
$$= \frac{1}{x} \left[\frac{\pi}{2} - \left(-\frac{\pi}{2}\right) \right]$$
$$= \frac{\pi}{x}$$

put this in eq (1)

$$= \frac{\rho_s x}{2\epsilon_0} \times \frac{\pi}{x}$$

$$E_x = \frac{\rho_s}{2\epsilon_0}$$

Finite sheet charge

$$E = \frac{Q}{4\pi\epsilon_0 R^2} \hat{a}_R$$

Point charge

$$E = \frac{Q}{2\pi\epsilon_0 R} \hat{a}_R$$

Line Charge

DAY-11

Sheet of Charge

Problem:-

Three uniform sheets of charge are located in free

Date

DAY-15

space as follows:-

- ① 3 nC/m^2 at $z = -4$
- ② 6 nC/m^2 at $z = 1$
- ③ -8 nC/m^2 at $z = 4$

Find E at point $P_A(2, 5, -5)$
at point $P_C(-1, -5, 2)$
at point $P_B(4, 2, -3)$
at point $P_D(-2, 4, 5)$

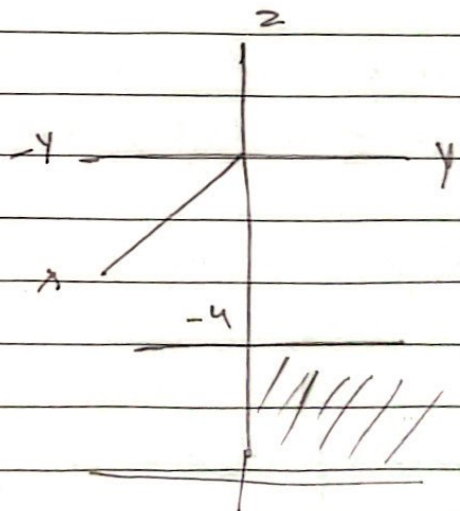
Sol:-

$$E = \frac{\rho_s}{2\epsilon_0} \cdot a_n$$

$$E_1 = 169.414 \text{ V/m}, \quad E_2 = 338.829 \text{ V/m}$$

$$E_3 = -451.97 \text{ V/m}$$

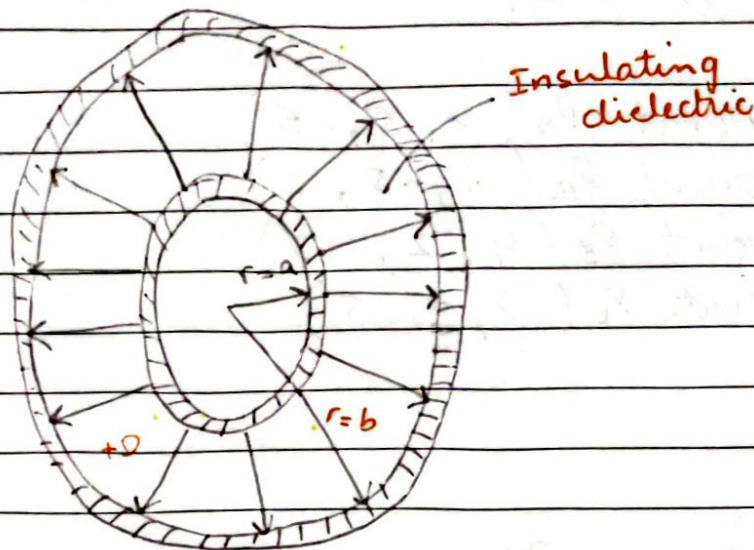
$$E_T = E_1 + E_2 + E_3$$
$$= 56.23 \text{ V/m}$$



DAY-12

Ch# 3

Electric Flux Density



→ Electric flux (Ψ) is directly proportional to charge on the inner sphere

$$\Psi = Q \quad (\text{Coulombs})$$

→ Electric flux density is defined as the electric flux per unit area.

→ Electric flux at any given point is given by the no. of flux lines crossing a surface normal to the lines divided by the surface.

→ Due to point charge, electric flux density is

$$D = \frac{\Psi}{4\pi r^2} \hat{a}_r = \frac{Q}{4\pi r^2} \hat{a}_r$$

the electric field intensity is

$$E = \frac{Q}{4\pi \epsilon_0 r^2} \quad ; \quad D = E \epsilon_0$$

Date

$$E = \int \frac{\rho dv}{4\pi\epsilon_0 r^2} \hat{a}_r$$

$$E_{\text{enc}} = \int \frac{\rho dv}{4\pi r^2} \hat{a}_r$$

$$D = \int \frac{\rho dv}{4\pi r^2} \hat{a}_r$$

QUESTION:-

Given a w.p.c point charge located at the origin, find the total electric flux passing through

a) the portion of the sphere $r = 26 \text{ cm}$ bounded by $0 < \theta < \frac{\pi}{2}$

and $0 < \phi < \pi$

Sol
 $\frac{1}{8}$ th section of the total sphere

$$\psi = 0$$

$$\psi = 0 \times \frac{1}{8}$$

$$= 60 \times \frac{1}{8}$$

$$= 7.5 \text{ nC}$$

b) Closed surface defined $\rho = 26 \text{ cm}$, $z \geq \pm 26 \text{ cm}$

$$\psi = 60 \text{ nC}$$

c) Plane $z = 26 \text{ cm}$

Date

QUESTION:-

Calculate D in RCS at point $P(2, -3, 6)$

a) Due to point charge $Q_A = 55 \text{ mC}$ at $Q(-2, 3, -6)$

$$D = \frac{Q}{4\pi r^2} \hat{r} \quad \text{Distance b/w } Q \text{ and } P$$

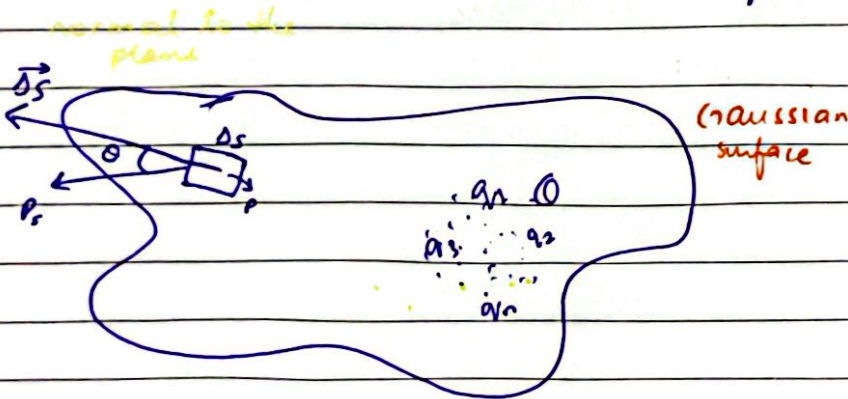
$$r = 14 \text{ m}$$

$$\hat{r} = \frac{-2}{7} a_x + \frac{3}{7} a_y - \frac{6}{7} a_z$$

DAY-13

Gauss's Law:-

The electric flux passing through any closed surface is equal to the total charge enclosed by the surface.



$$\Delta \Psi = \text{flux passing through } \Delta S$$

$$= D_s \Delta S \cos \theta$$

$$\text{Total flux} = \Psi = \int d\Psi = \int D_s \cdot \Delta S = \oint_{\text{closed surface}} D_s \cdot dS = Q$$

closed surface integral

Date

$$\text{RCS} \rightarrow \Psi = Q = \int_y \int_x D(x,y) dx dy$$

$$\text{CCS} \rightarrow \Psi = Q = \int_{\phi} \int_{\rho} D_{\phi} \rho d\phi d\rho$$

$$\text{SCS} \rightarrow \Psi = Q = \int \int D_{\theta} r^2 \sin\theta d\theta d\phi$$

Point charges :

$$\Psi = Q = q_1 + q_2 + q_3 + \dots + q_n$$

$$\Sigma Q_n = \oint D_s \cdot ds$$

Line charge ::

$$\Psi = Q = \int P_L dL = \oint D_s \cdot ds$$

Surface charge ::

$$\Psi = Q = \int P_L dS = \oint D_s \cdot ds$$

Volume charge distribution ::

$$\Psi = Q = \int P_V dv = \oint_s D_s \cdot ds$$

QUESTION ::

Given

PI-VAD

$D = 0.3r^2 a_r \text{ nC/m}^2$ in free space

- ① Find E at point P ($r=2$, $\theta=25^\circ$, $\phi=90^\circ$)
- ② Find the total charge within sphere $r=3$
- ③ Find the total electric flux leaving sphere $r=4$

SOL :-

$$a) \quad E = \frac{Q}{4\pi\epsilon_0 r^2} a_r$$

Date

 $D = E \epsilon_0 \rightarrow$ using this formula

$$D = \frac{Q}{4\pi r^2} ar$$

$$E = \frac{D}{\epsilon_0}$$

$$= \frac{0.3 \times 10^{-9} r^2 ar}{8.85 \times 10^{-12}}$$

$$\boxed{E = 135.6 ar \text{ V/m}}$$

$$b) D = \frac{Q}{4\pi r^2} ar$$

$$4\pi r^2 \times 0.3 r^2 ar = Q$$

$$Q = 4\pi (3)^2 \times (0.3 \times 10^{-9}) (3)^2$$

$$\boxed{Q = 305 \times 10^{-9} \text{ C}}$$

$$c) D = \frac{Q}{4\pi r^2} ar$$

$$Q = 965 \times 10^{-9} \text{ C}$$

$$\boxed{Q = \gamma = 965 \text{ nC}}$$

DAY-14

Applications of Gauss's Law Symmetrical Charge Distributions :-

$\oint D \cdot ds = Q \rightarrow$ only applicable for symmetrical surfaces

symmetry :: (spherical, cylindrical, planar symmetrical)

Key conditions for applying Gauss's law ::

To effectively use Gauss's law, we must choose

Date

a Gaussian surface that satisfies the following condition:

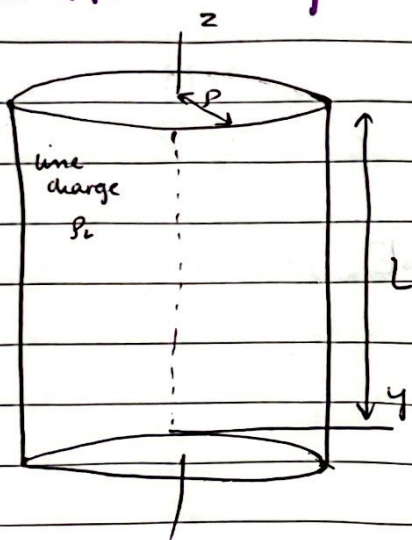
- ① D (Electric Flux Density) must be either normal or tangential to the surface $\rightarrow$ If D is normal to the surface $D \cdot ds = D ds$
 $\rightarrow$ If D is tangential to the surface $D \cdot ds = 0$
- ② The magnitude of D must be constant over the surface where it is normal

— this allows us to pull D outside of the integral, leaving only the surface area inside the integral.

For cylindrical surface & point charge:-

Gaussian surface
of an infinite
large charge

D is constant in
magnitude where as
perpendicular to the
cylindrical surface



$$\begin{aligned}\Phi &= \oint D_s \cdot ds = D_s \int ds \\ &= D_s \int_{\text{top}} ds + D_s \int_{\text{bottom}} ds + D_s \int_{\text{sides}} ds \\ &= 0 \int_{\text{top}} ds + 0 \int_{\text{bottom}} ds + D_s \int_{\text{sides}} ds \\ &= D_s \int_{z=0}^L \int_{\phi=0}^{2\pi} r d\phi dz \\ &= D_s 2\pi r L\end{aligned}$$

Date

$$Q = D_s 2\pi p L$$

$$D_s = \frac{Q}{2\pi p L}$$

$$D_s = D_s = \frac{Q}{2\pi p L}$$

Also for line charge

$$Q = p L$$

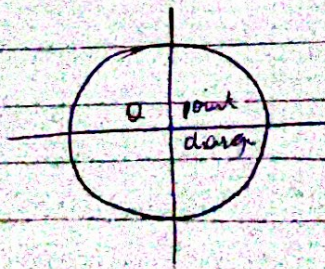
then

$$D_s = D_p = \frac{p L}{2\pi p L}$$

$$D_p = \frac{p}{2\pi p}$$

$$E = \frac{p}{2\pi \epsilon_0 p} \quad \text{because } D = \epsilon_0 E$$

For sphere & point charge :-



$$Q = \oint D_s ds$$

$$Q = D_s \oint ds$$

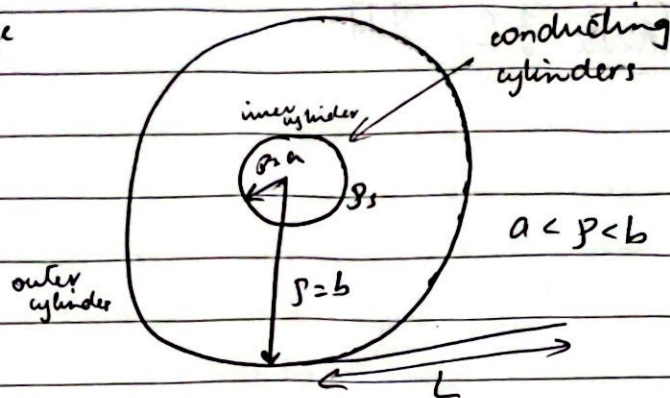
$$Q = D_s 4\pi r^2$$

Date

$$D_s = \frac{Q}{4\pi r^2} \hat{a}_r$$

$$E = \frac{Q}{4\pi \epsilon_0 r^2} \hat{a}_r$$

ρ_s = surface charge density



Coaxial cable with cylindrical symmetry :-

$$D_s = D_p = \frac{Q}{2\pi \rho L} \Rightarrow Q = D_s 2\pi \rho L$$

the total charge on a length L of the inner conductor is

$$Q = \int_{z=0}^L \int_{\phi=0}^{2\pi} \rho_s a d\phi dz = 2\pi a L \rho_s = Q$$

$$Q = Q$$

$$2\pi a L \rho_s = D_s 2\pi \rho L$$

$$D_s' = \frac{a \rho_s}{\rho} \hat{a}_r$$

$$Q_{enc \text{ inner}} = 2\pi a L \rho_s$$

$$Q_{enc \text{ outer}} = -2\pi a L \rho_{s \text{ inner}}$$

Date

$$Q_{\text{enc outer}} = -Q_{\text{enc inner}}$$

$$\cancel{2\pi b} \cancel{L} \epsilon_0 = -\cancel{2\pi a} \cancel{L} \epsilon_0 \cancel{b_{\text{inner}}}$$

$$\frac{\epsilon_0 a_{\text{outer}}}{\epsilon_0 b_{\text{inner}}} = -\frac{a}{b}$$

Example 3.2 H.W.